

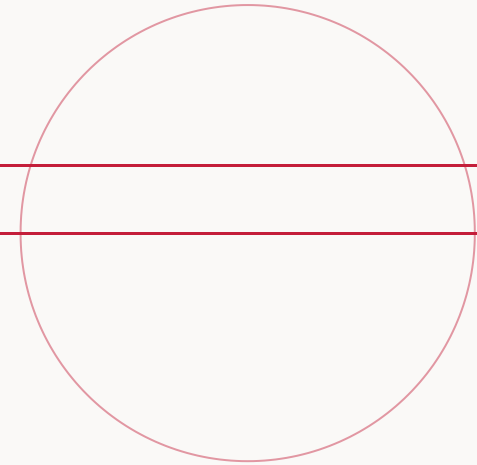
Statistical Estimation & Inference

*How confident can we really be
in the numbers we compute?*

Manaranjan Pradhan

www.aithinking.in

A Course on Quantitative Finance



MOTIVATION

Every estimate carries a hidden second number

We have spent two modules computing means, standard deviations and quantiles — point estimates of a stock's *expected return* and *risk*. Each comes from one finite sample, so each is just an *estimate*, not the truth.

- How precise is $\bar{r}$ — the estimated return?
- How precise is s — the estimated risk?
- Could the true mean plausibly be **zero**?
- Are two stocks' means **really** different, or sampling noise?

THE EXPERIMENT

Two paths to a weekly return

One year of daily prices for SBI and Yes Bank. We will deliberately construct weekly returns in two different ways — one honest, one tempting but treacherous.

METHOD A · HONEST

Friday-close weekly

$$n = 51$$

One return per week. Non-overlapping. Effectively iid.

METHOD B · TEMPTING

5-day rolling weekly

$$n = 237$$

A weekly return for every day. Heavily overlapping.

Both target the same quantity. 5× the observations — but is it 5× the information?

CHAPTER I

The honest path
— *Friday-close weekly*
returns

POINT ESTIMATORS

Sample mean and sample standard deviation

From the sample $r_1, \dots, r_n$, the standard estimators of the true mean and variance are:

UNBIASED POINT ESTIMATORS

$$\hat{\mu} = \bar{r} = \frac{1}{n} \sum_{i=1}^n r_i$$

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2$$

The $n-1$ in the denominator makes s^2 unbiased — $E[s^2] = \sigma^2$.

These are **numbers** — one per dataset. The truth is a parameter we never observe.

FRIDAY-CLOSE ESTIMATES

One year, two stocks, four numbers

SBI

 $\bar{r} \cdot s$

+0.690%

weekly mean · std = 2.96%

YES BANK

 $\bar{r} \cdot s$

+0.299%

weekly mean · std = 4.16%

*Yes Bank's weekly volatility is ~40% **larger** than SBI's, but its mean return is less than half. Is the difference real, or is one year just one year?*

SAMPLING VARIABILITY

Standard error of the mean

If we collected a different year, we would get a different $\bar{r}$. The standard deviation of $\bar{r}$ across hypothetical resamples is its **standard error**.

STANDARD ERROR OF THE SAMPLE MEAN

$$\text{SE}(\bar{r}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

Replace the unknown σ with its sample estimate s . SE shrinks at rate $1/\sqrt{n}$.

For SBI: **SE = 2.96% / $\sqrt{51} \approx 0.415\%$** . Yes Bank: **SE = 4.16% / $\sqrt{51} \approx 0.583\%$** .

A PIVOTAL DISTINCTION

Sample distribution vs *sampling* distribution

One letter apart, two different objects. Confusing them is the single most common source of error in applied statistics.

SAMPLE DISTRIBUTION

$$\text{spread} = \sigma$$

$$r_1, \dots, r_n$$

Histogram of the n observed returns themselves.

SAMPLING DIST. OF $\bar{r}$

$$\text{spread} = \sigma/\sqrt{n}$$

$$\bar{r} \text{ across resamples}$$

Distribution of the statistic, not the data.

Central Limit Theorem — as n grows, $\bar{r} \rightarrow N(\mu, \sigma^2/n)$ regardless of the shape of the sample distribution. That is why $s/\sqrt{n}$ works even when returns are fat-tailed.

A MODEL-FREE ALTERNATIVE

The bootstrap

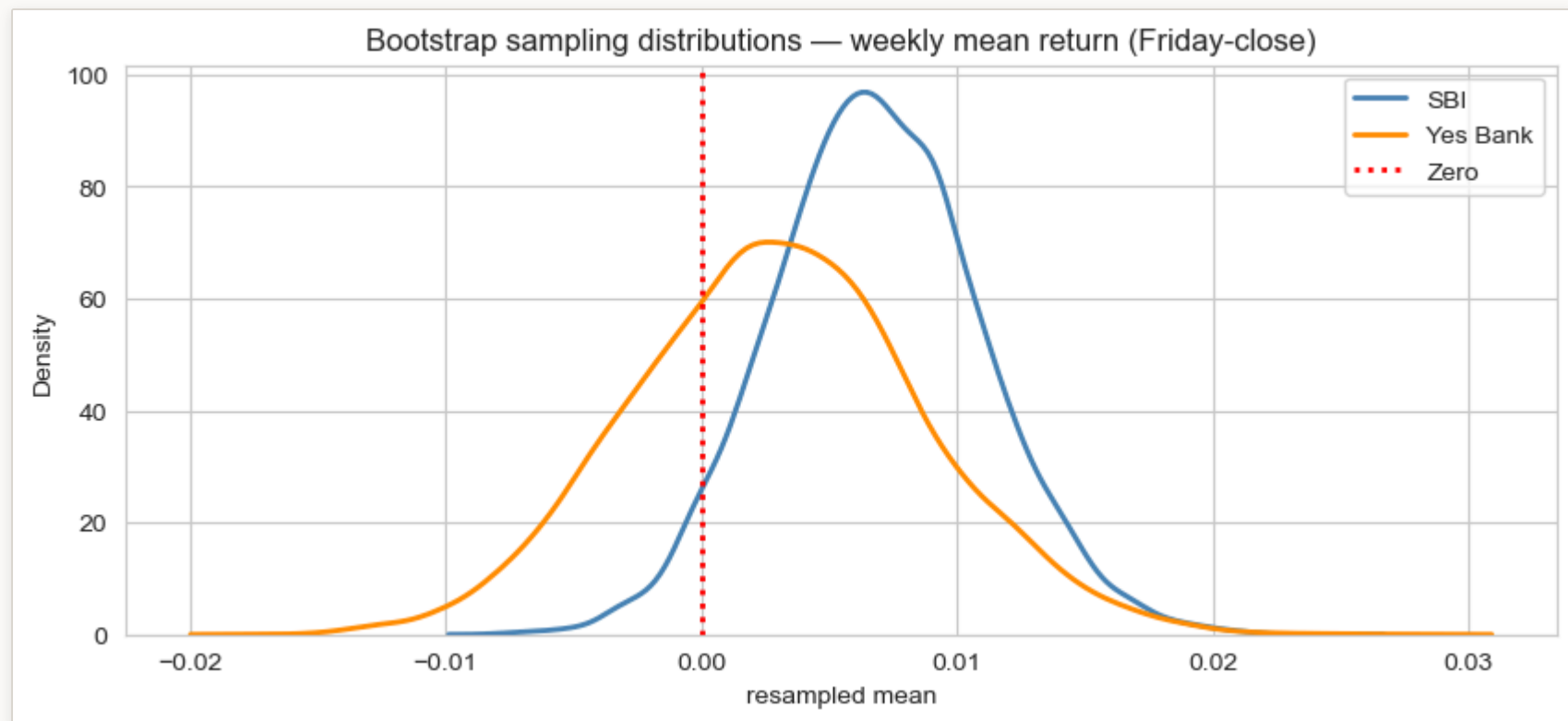
Don't trust the $s/\sqrt{n}$ formula? Simulate the sampling distribution directly.

- **Resample** n values from the data *with replacement*.
- Compute the **mean** of that resample.
- **Repeat** 10,000 times.
- The standard deviation of the resampled means is the **bootstrap SE**.

SBI: *analytical SE = 0.415% · bootstrap SE = 0.413%. Close agreement — the formula behaves on this data.*

Bootstrap sampling distributions of $\bar{r}$

- ▶ Both distributions are **roughly bell-shaped**
- ▶ Yes Bank's is **wider** — larger s , larger SE
- ▶ Both **straddle zero** — the mean is uncertain



QUANTIFYING THE UNCERTAINTY

95% confidence interval

Three constructions, all aiming at the same target: an interval that would cover the true μ in 95% of repeated samples.

THREE CONSTRUCTIONS

(a) Normal: $\bar{r} \pm 1.96 \cdot \text{SE}$

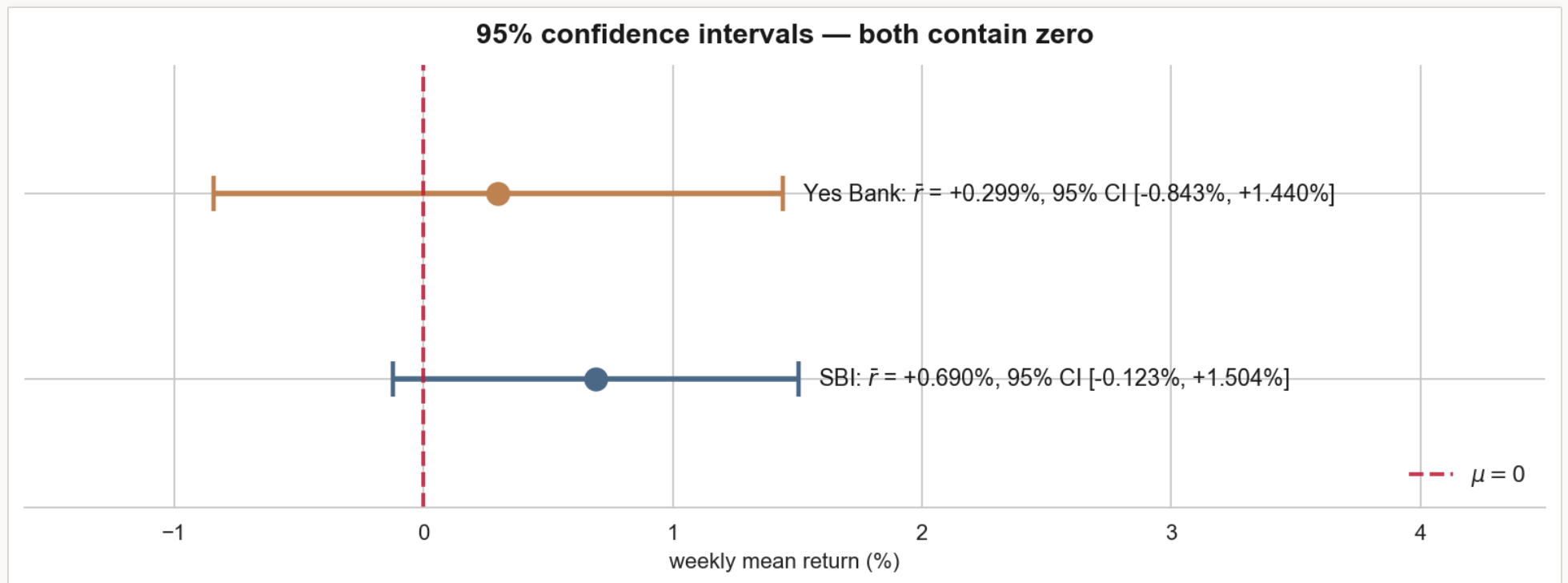
(b) t-CI: $\bar{r} \pm t_{0.975, n-1} \cdot \text{SE}$

(c) Bootstrap: $[Q_{2.5}(\bar{r}^*), Q_{97.5}(\bar{r}^*)]$

For $n = 51$, the t -multiplier 2.009 is barely larger than 1.96 — all three intervals are close.

Both intervals contain zero

- ▶ SBI CI: $[-0.12\%, +1.50\%]$
- ▶ Yes Bank CI: $[-0.84\%, +1.44\%]$
- ▶ One year of weekly data **cannot rule out $\mu = 0$**



HYPOTHESIS TESTING

One-sample t-test — $H_0: \mu = 0$

Five ingredients: the null, the alternative, a test statistic, a p-value, and a decision rule.

TEST STATISTIC AND DECISION

$$t = \frac{\bar{r} - 0}{s/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0$$

Reject H_0 if $p < \alpha$ (conventionally 0.05)

Two-sided alternative $\mu \neq 0$. p -value = $P(|T| \geq |t|)$ under H_0 .

RESULTS · ONE-SAMPLE T-TEST

Neither stock rejects $H_0: \mu = 0$

	$\bar{R}$ (WEEKLY)	T	P	DECISION @ $\alpha = 0.05$
SBI	+0.690%	+1.66	0.102	fail to reject
Yes Bank	+0.299%	+0.51	0.610	fail to reject

Fail to reject $\neq$ accept. We have not proved $\mu = 0$; we have only failed to find enough evidence against it in 51 observations.

TWO STOCKS, ONE QUESTION

Welch's two-sample t-test

Are SBI and Yes Bank's mean weekly returns *really* different?

Welch's variant does not assume equal variances — ideal here.

WELCH T-STATISTIC — $H_0: M_1 = M_2$

$$t = \frac{\bar{r}_1 - \bar{r}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

Observed difference: $+0.690\% - 0.299\% = +0.392\%$. $t = +0.55$, $p = 0.585$.

Fail to reject. The 0.4% difference looks meaningful in a brochure, but with this much noise it is well within sampling variability.

CHAPTER II

The trap — 5-day rolling weekly returns

ROLLING WEEKLY RETURNS

Same means, same vols — five times the n

Compute $r_t = (P_t - P_{t-5})/P_{t-5}$ for every trading day. Each consecutive observation overlaps the last in 4 of 5 underlying days.

SBI · ROLLING

 $n = 237$ **+0.731%**

mean · std = 2.94%

YES BANK · ROLLING

 $n = 237$ **+0.300%**

mean · std = 4.11%

*Means and vols match Section A — only n changed. Naïvely, $SE = s/\sqrt{237}$ would be $\approx$ **0.19%**, less than half the honest SE.*

WHY THE FORMULA BREAKS

The $s/\sqrt{n}$ formula assumes independence

$SE = \sigma/\sqrt{n}$ is a consequence of $Var(\bar{r}) = \sigma^2/n$ — which itself assumes the observations are **independent**.

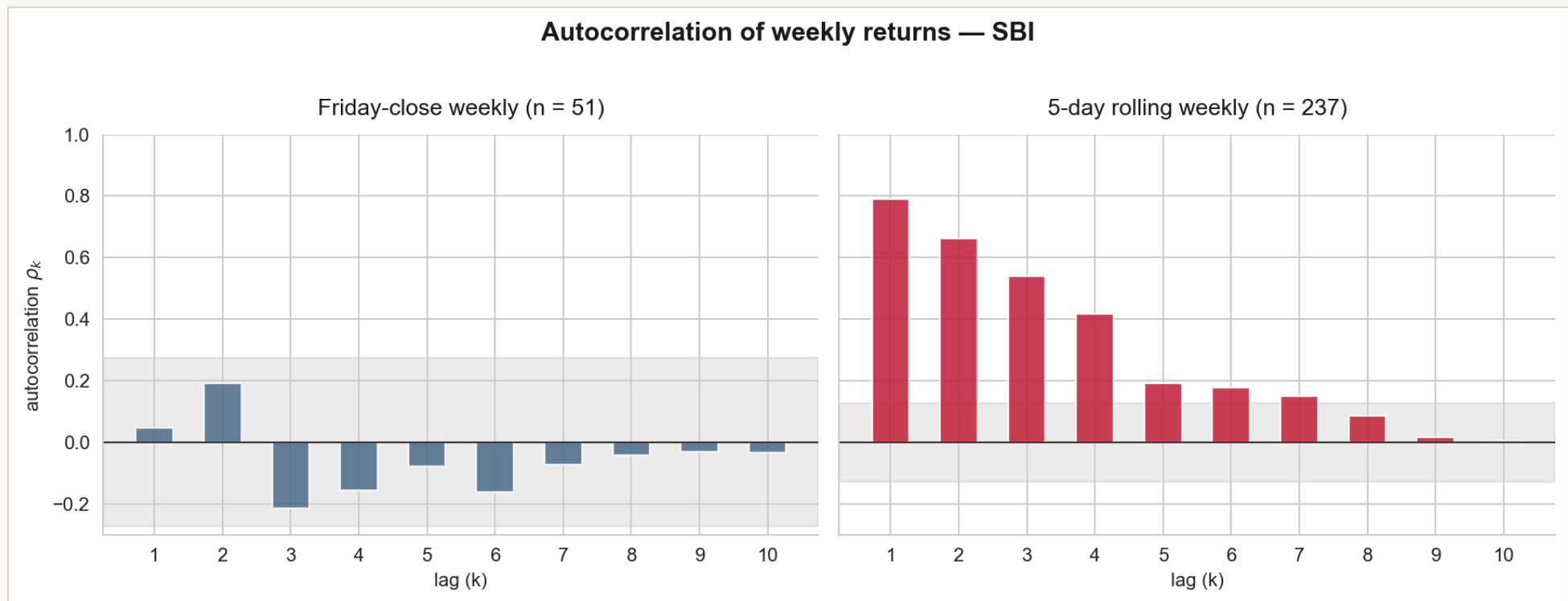
- Today's 5-day return shares **4 days** with yesterday's.
- The series is **strongly autocorrelated** by construction.
- $n = 237$ data points carry **far less** than 237 units of information.
- Apply the iid formula anyway and you get **fake precision**.

Autocorrelation exposes the overlap

► Friday-close: **noise** — bars sit inside the band

► Rolling: lag-1 ≈ 0.79 , decays slowly to lag 5

► Exactly the **4-of-5 overlap**, made visible

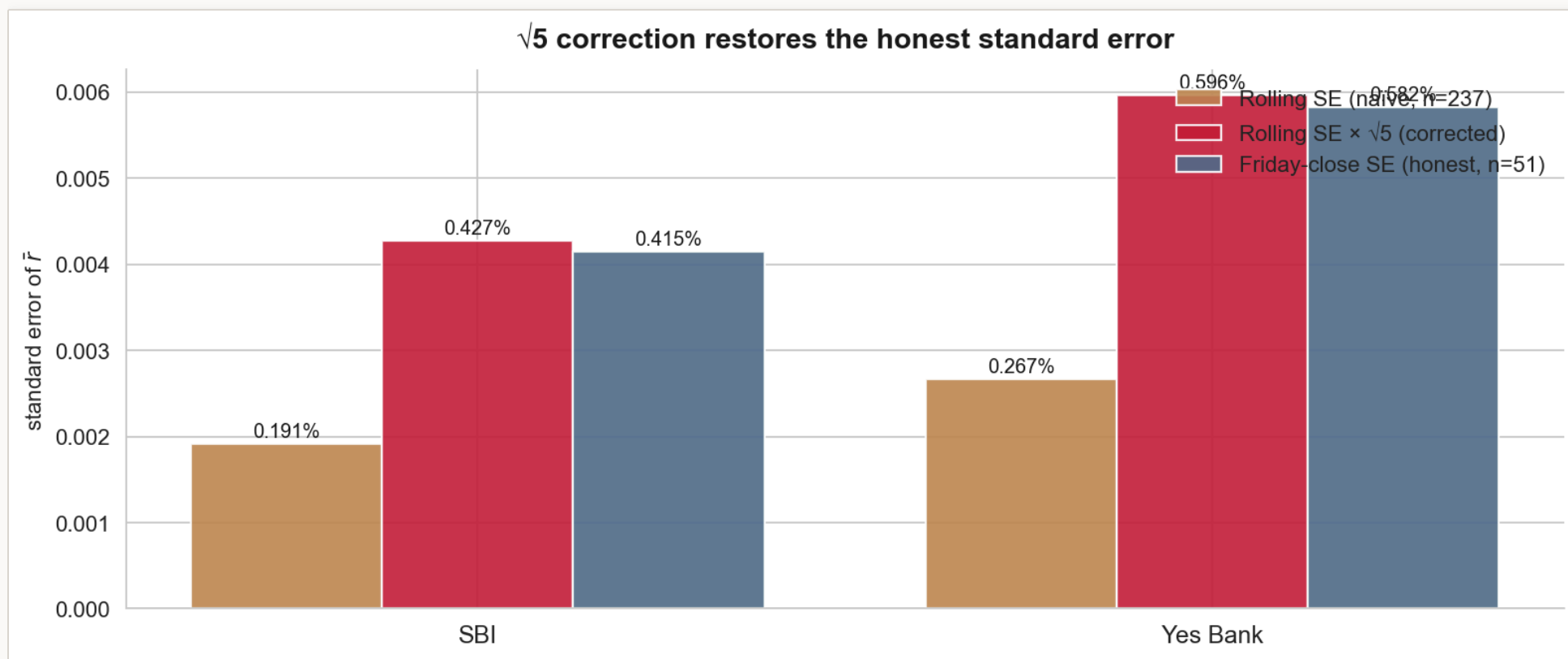


More numbers $\neq$ more information

► Naïve rolling SE looks $\sim\frac{1}{2}$ the honest SE

► Inflate by $\sqrt{5}$ for the 5-day overlap

► Result: **matches** the Friday-close SE



CHAPTER III

A test of a different shape
— *Kolmogorov–Smirnov*

GOODNESS OF FIT

From a parameter to a whole distribution

The t-test asked about a single number, μ . The K-S test asks whether the *entire CDF* matches a hypothesised distribution.

KOLMOGOROV-SMIRNOV STATISTIC

$$D = \sup_x |F_n(x) - F(x)|$$

Largest vertical gap between the empirical CDF and the fitted CDF. Same decision rule: reject H_0 if $p < 0.05$.

A **much stronger claim** than $\mu = 0$. K-S can reject even when the t-test on the mean does not.

RESULTS · K-S ON WEEKLY RETURNS

Normal is plausible at the weekly horizon

	N	D	P	DECISION @ $\alpha = 0.05$
SBI	51	0.128	0.341	fail to reject
Yes Bank	51	0.122	0.403	fail to reject

Aggregation tames the tails. *Module 2 found Normal a poor fit for daily returns. By the weekly horizon, with only 51 points, K-S cannot reject Normal.*

CLOSING

Five things to remember

- Every estimate has a **standard error** — a point estimate without it is half a number.
- The **bootstrap** and the **analytical SE** agree on iid data — useful when one breaks down.
- One year of weekly returns **cannot reject $\mu = 0$** , for either stock or for their difference.
- **Overlapping observations** inflate n but not information — the iid SE understates uncertainty.
- The **$\sqrt{5}$ correction** on rolling weekly SE returns it to the honest Friday-close value.

More numbers do not mean more information.

— ask what is independent, not what is large.